

# Executive Summary

## **Project: Factory Reallocation & Shipping Optimization Recommendation System**

Nassau Candy Distributor operates multiple factories responsible for manufacturing and distributing confectionery products across different regions. Existing product-to-factory assignments are largely based on static business rules, which can lead to inefficiencies in production planning, shipping operations, and resource utilization.

This project developed a data-driven Factory Reallocation and Shipping Optimization Recommendation System to support better operational decision-making. Using historical sales, profitability, demand, and shipping data from more than 10,000 business records, the system identifies opportunities to improve factory assignments and overall supply chain performance.

Key findings from the analysis indicate that Lot's O' Nuts and Wicked Choccy's are the highest-performing factories, while the Pacific region contributes the largest share of profitability. Several Wonka product variants were identified as high-demand, high-profit products that should remain prioritized within production planning strategies.

To support decision-makers, an interactive Streamlit dashboard was developed with the following capabilities:

- Factory profitability analysis
- Regional performance monitoring
- Product demand and profitability analysis
- Product reassignment recommendations
- Factory optimization simulation
- What-if scenario analysis
- Risk and impact assessment
- Interactive filtering and reporting

The proposed solution enables stakeholders to evaluate the potential impact of factory reassignment decisions before implementation, reducing operational risks and improving resource allocation. The recommendation system supports evidence-based planning and can help increase profitability while maintaining service quality and operational efficiency.

## **Key Benefits**

- Improved factory utilization and production planning.
- Better alignment between product demand and manufacturing capacity.
- Identification of high-profit products and regions.
- Enhanced decision support through interactive analytics.
- Reduced risk associated with factory reassignment decisions.

## **Strategic Impact**

The project demonstrates how data analytics can be applied to supply chain and manufacturing operations to support smarter allocation decisions. The framework can be expanded in the future through machine learning, demand forecasting, lead-time prediction, and transportation route optimization, creating a foundation for more advanced operational intelligence systems.

Overall, the proposed system provides a practical and scalable approach for improving manufacturing efficiency, profitability, and decision-making within Nassau Candy Distributor's operations.